EECS 16A: Introduction to Signals, Dynamics, and Information Exam 2

Department of Electrical Engineering and Computer Sciences

University of California, Berkeley 19 March 2026

FIRST Name: Larry LAST Name: PaigeRank SID (All Digits): 0123456789

- **(5 Points)** On *every* page, print legibly your name and ALL digits of your SID. For every page on which you do not write your name and SID, you forfeit a point, up to the maximum 5 points.
- **(10 Points) (Pledge of Academic Integrity)** Hand-copy, sign, and date the single-line text (which begins with *I have read, . . .*) of the Pledge of Academic Integrity on page 3 of this document. Your solutions will *not* be evaluated without this.
- **Urgent Contact with the Teaching Staff:** In case of an urgent matter, raise your hand.
- **This document consists of pages numbered 1 through 12.** Verify that your copy of the exam is free of anomalies, and contains all of the specified number of pages. If you find a defect in your copy, contact the teaching staff immediately.
- You have 80 minutes total—*in one sitting*—to tackle this exam.

  The exam starts at 8:10 pm California time. Your allotted window begins with respect to this start time. Students who have official accommodations of 1.5× and 2× time windows have 120 and 160 minutes, respectively.
- **This exam is closed book.** You may not use or access, or cause to be used or accessed, any reference in print or electronic form at any time during the exam, except two double-sided 8.5" × 11" sheet of handwritten, original notes having no appendage.

  Collaboration is not permitted.

  Computing, communication, and other electronic devices (except dedicated timekeepers) must be turned off.

  Scratch paper will be provided to you; ask for more if you run out. You may not use your own scratch paper.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it.*
- For each problem, limit your work to the space provided specifically for that problem. *No other work will be considered. For example, we will not evaluate scratch work. No exceptions.*
- Unless explicitly waived by the specific wording of a problem, you must explain your responses (and reasoning) succinctly, but clearly and convincingly.
- In some parts of a problem, we may ask you to establish a certain result—for example, "show this" or "prove that." Even if you're unable to establish the result that we ask of you, you may still take that result for granted—and use it in any subsequent part of the problem.

FIRST Name: Larry LAST Name: PaigeRank SID (All Digits): 0123456789

- If we ask you to provide a "reasonably simple expression" for something, by default we expect your expression to be in closed form—one *not* involving a sum $\sum$ or an integral $\int$—*unless* we explicitly tell you otherwise.
- Noncompliance with these or other instructions from the teaching staff—*including, for example, commencing work prematurely, or continuing it beyond the allocated time window*—is a serious violation of the Code of Student Conduct.

FIRST Name: Larry LAST Name: PaigeRank SID (All Digits): 0123456789

Pledge of Academic Integrity

By my honor, I affirm that

(1) this document—which I have produced for the evaluation of my performance—reflects my original, bona fide work, and that I have neither provided to, nor received from, anyone excessive or unreasonable assistance that produces unfair advantage for me or for any of my peers;

(2) as a member of the UC Berkeley community, I have acted with honesty, integrity, respect for others, and professional responsibility—and in a manner consistent with the letter and intent of the campus Code of Student Conduct;

(3) I have not violated—nor aided or abetted anyone else to violate—the instructions for this exam given by the course staff, including, but not limited to, those on the cover page of this document; and

(4) More generally, I have not committed any act that violates—nor aided or abetted anyone else to violate—UC Berkeley, state, or Federal regulations, during this exam.

(10 Points) In the space below, hand-write the following sentence, verbatim. Then write your name in legible letters, sign, include your full SID, and date before submitting your work:

I have read, I understand, and I commit to adhere to the letter and spirit of the pledge above.

I have read, I understand, and I commit to the letter and spirit of the pledge above

Full Name: Larry PaigeRank Signature: L

Date: 19 March 2026 Student ID: 0123456789

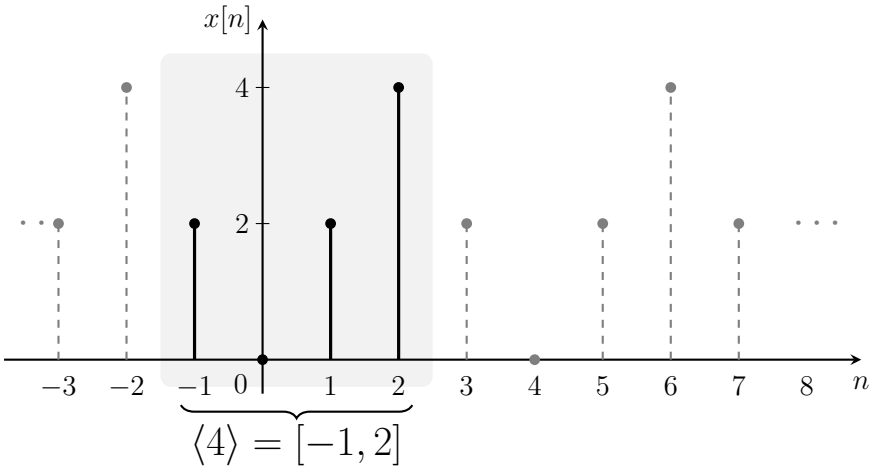
FIRST Name: Larry LAST Name: PaigeLank SID (All Digits): 0123456789

E2.1 (45 Points) DTFS Bonanza!

Consider a 4-periodic discrete-time (DT) signal x described over one of its periods $\langle 4 \rangle = [-1, 2]$ as follows:

$$x[n] = \begin{cases} 2 & n = -1, 1 \\ 0 & n = 0 \\ 4 & n = 2 \end{cases},$$

It's understood that the pattern repeats 4-periodically outside the interval described. The figure below depicts a stem plot of this signal, with the principal period highlighted.



$P = 4$

$\omega_0 = \frac{2\pi}{P} = \frac{\pi}{2}$

(a) (30 Points) Determine a reasonably simple expression for the DTFS coefficients X_k, $k = -1, 0, 1, 2$, and sketch (on the figure provided near the top of the next page) a well-labeled stem plot for X_k for $k \in [-4, 3]$.

$$X_k = \frac{1}{4} \sum_{n=-1}^{2} x[n] e^{-ik\frac{\pi}{2}n} = \frac{x[-1]e^{ik\pi/2} + \cancel{x[0]}^{0} + x[1]e^{-ik\pi/2} + x[2]e^{-ik\pi}}{4}$$

$$X_k = \frac{e^{ik\pi/2} + e^{-ik\pi/2}}{2} + e^{-ik\pi} \Rightarrow$$

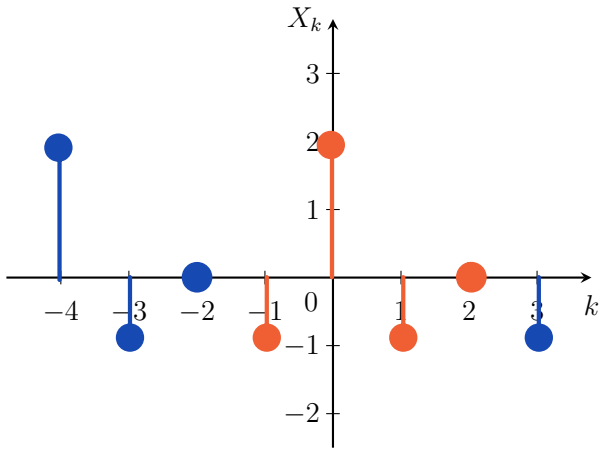
$$\boxed{X_k = \cos\left(\frac{k\pi}{2}\right) + (-1)^k}$$

$$X_{-1} = X_3 = \cos\left(\frac{3\pi}{2}\right) + (-1)^3 = -1 \Rightarrow X_{-1} = X_3 = -1 \ldots$$

$$X_0 = X_{-4} = \cos 0 + (-1)^0 = 2 \Rightarrow X_0 = X_{-4} = 2$$

$$X_1 = X_{-3} = \cos\left(\frac{\pi}{2}\right) + (-1)^1 = -1 \qquad X_2 = X_{-2} = \cos\pi + (-1)^2 = 0$$

E2.1 (Continued)



(b) (15 Points) You're told that you may save only three of the four DTFS coefficients X_{-1}, X_0, X_1, X_2 from which to reconstruct an estimate $\widehat{x}$ of the signal x.

Which three coefficients would you keep? Explain your choice. And for your choice, determine the numerical value of the squared error

$$\|\varepsilon\|^2 = \sum_{n=-1}^{2} |\varepsilon[n]|^2 = \sum_{n=-1}^{2} \left|x[n] - \widehat{x}[n]\right|^2.$$

We'll keep the three nonzero coefficients: X_{-1}, X_0, and X_1.

The error would be zero, since $x[n]$ is completely determined by the three terms associated with X_{-1}, X_0, and X_1.

FIRST Name: Larry LAST Name: PaigeRank SID (All Digits): 0123456789

E2.2 (40 Points) DTFS Optimality

Consider a p-periodic DT signal $x : \mathbb{Z} \to \mathbb{R}$ whose DTFS expansion is given by $x[n] = \sum_{k=0}^{p-1} X_k \psi_k[n]$, where $\psi_k[n] = e^{ik\omega_0 n}$ are the complex-exponential basis signals, and $\omega_0 = 2\pi/p$ is the fundamental frequency of the signal.

We want to approximate the signal x with fewer than p terms—namely, with

$$\widehat{x}[n] = \sum_{k=0}^{N-1} \alpha_k \psi_k[n], \qquad \text{for some integer } N < p.$$

(a) (25 Points) Show that the coefficients α_k that minimize the squared error

$$\|\varepsilon\|^2 = \sum_{n=0}^{p-1} |\varepsilon[n]|^2 = \sum_{n=0}^{p-1} \left|x[n] - \widehat{x}[n]\right|^2$$

are, in fact, the DTFS coefficients—that is, the optimal coefficients are

$$\alpha_k = X_k, \quad k = 0, 1, \ldots, N-1.$$

The squared error is minimized by ensuring that the error signal $\varepsilon = x - \hat{x}$ is orthogonal to each of the basis signals ψ_l, $l = 0, 1, \ldots, N-1$.

$$\langle \varepsilon, \psi_k \rangle = \langle x - \hat{x}, \psi_k \rangle = \langle x, \psi_k \rangle - \langle \hat{x}, \psi_k \rangle = 0$$

$$\Rightarrow \langle \sum_{l=0}^{N-1} \alpha_l \psi_l, \psi_k \rangle = \langle x, \psi_k \rangle \Rightarrow$$

$$\sum_{l=0}^{N-1} \alpha_l \langle \psi_l, \psi_k \rangle = \alpha_k \langle \psi_k, \psi_k \rangle = \langle x, \psi_k \rangle \Rightarrow$$

b/c $\psi_l \perp \psi_k$ $k \neq l$

$\alpha_k = \dfrac{\langle x, \psi_k \rangle}{\langle \psi_k, \psi_k \rangle}$. But the right-hand side is precisely $X_k \Rightarrow$ $\boxed{\alpha_k = X_k}$ $k = 0, \ldots, N-1$

The optimal coefficients are the DTFS coeffs.

FIRST Name: Larry LAST Name: PaigeRank SID (All Digits): 0123456789

E2.2 (Continued)

(b) (15 Points) Determine a reasonably simple expression for the squared error $\|\varepsilon\|^2$ in terms of the DTFS coefficients X_k.

Your expression should be in a summation form (not closed form), and should involve some combination of X_k, p, and/or ω_0.

Since $\alpha_k = X_k \quad k = 0, 1, \ldots, N-1 \Rightarrow$

$$\varepsilon = x - \hat{x} = \sum_{k=N}^{p-1} X_k \psi_k \Rightarrow$$

$$\langle \varepsilon, \varepsilon \rangle = \left\langle \sum_{k=N}^{p-1} X_k \psi_k, \sum_{l=N}^{p-1} X_l \psi_l \right\rangle$$

$$= \sum_{k=N}^{p-1} \sum_{l=N}^{p-1} X_k X_l^* \langle \psi_k, \psi_l \rangle \Rightarrow$$

But $\langle \psi_k, \psi_l \rangle = p\,\delta[k-l]$

$$\langle \varepsilon, \varepsilon \rangle = p \sum_{k=N}^{p-1} X_k X_k^* \Rightarrow \|\varepsilon\|^2 = p \sum_{k=N}^{p-1} |X_k|^2$$

FIRST Name: Larry LAST Name: PaigeRank SID (All Digits): 0123456789

E2.3 (35 Points) DTFS Properties

Consider a *real-valued*, p-periodic DT signal x whose DTFS expansion is given by $x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n}$, where $\omega_0 = 2\pi/p$ is the fundamental frequency.

We construct a new signal y as follows: $y[n] = \dfrac{x[n] + x[-n]}{2}$ for all integers n.

The signal y is called the *even part of* the signal x. It's not hard to verify that $y[n] = y[-n]$.

(a) (15 Points) Show that the signal y—the even part of x—is p-periodic. That is, show that

$$y[n+p] = y[n] \qquad \forall n \in \mathbb{Z}.$$

We already know that $x[n+p]=x[n]$, $\forall n\in\mathbb{Z}$.
If we show that $x_{TR}[n+p]=x_{TR}[n]$, where x_{TR} is the time-reversed version of x — that is, $x_{TR}[n]=x[-n]$, $\forall n\in\mathbb{Z}$ — then we will have shown that $y[n+p]=y[n]$, $\forall n\in\mathbb{Z}$.

$x[n+p]=x[n]$ $\forall n\in\mathbb{Z}$

Let $l=-n-p \Rightarrow \begin{cases} n=-l-p=-(l+p) \\ n+p=-l \end{cases} \Rightarrow x[-l]=x[-(l+p)]$ $\forall l\in\mathbb{Z}$

$x[-l]=x_{TR}[l]$, $x[-(l+p)]=x_{TR}[l+p]$

So, we've shown that $x_{TR}[l+p]=x_{TR}[l]$, $\forall l\in\mathbb{Z}$ — that is, x_{TR} is p-periodic. Since $y[n]=\dfrac{x[n]+x_{TR}[n]}{2}$ and each term in the numerator corresponds to a p-periodic signal (x and x_{TR}), y is p-periodic.

E2.3 (Continued)

(b) (20 Points) Determine a reasonably simple expression for the DTFS coefficients Y_k in terms of X_k.

$$y = \frac{x + x_{TR}}{2} \Rightarrow Y_k = \frac{X_k + X_{TR,k}}{2}$$

Let's look at $X_{TR,k}$:

$$X_{TR,k} = \frac{1}{P}\sum_{n\in\langle P\rangle} x_{TR}[n]e^{-ik\omega_0 n} = \frac{1}{P}\sum_{n\in\langle P\rangle} x[-n]e^{-ik\omega_0 n} \Rightarrow$$

$x_{TR}[n] = x[-n]$

Let $m = -n \Rightarrow m \in \langle P\rangle$

$$X_{TR,k} = \frac{1}{P}\sum_{m\in\langle P\rangle} x[m]e^{ik\omega_0 m} = \frac{1}{P}\sum_{m\in\langle P\rangle} x^*[m]\left(e^{-ik\omega_0 m}\right)^*$$

$x[m] \in \mathbb{R}\ \forall m \in \mathbb{Z} \Rightarrow x^*[m] = x[m]\ \forall m \in \mathbb{Z}$

$$X_{TR,k} = \left(\underbrace{\frac{1}{P}\sum_{m\in\langle P\rangle} x[m]e^{-ik\omega_0 m}}_{X_k}\right)^* = X_k^*$$

So, we have

$$Y_k = \frac{X_k + X_k^*}{2} = \frac{2\,\mathrm{Re}(X_k)}{2} \Rightarrow \boxed{Y_k = \mathrm{Re}(X_k)}$$

If you argue that $X_{TR,k} = X_{-k}$ and leave your answer as $Y_k = \frac{X_k + X_{-k}}{2}$, you'll get partial credit, because it means that you have not taken advantage of x being real-valued.

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E2.4 (65 Points) PageRank Networks

The connectivity matrix of a PageRank network is shown below:

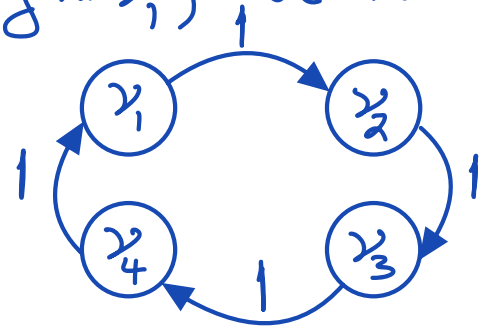
$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

The eigenvalues of this matrix are $\lambda_k = e^{ik\pi/2} = i^k$, $k = 0, 1, 2, 3$. That is,

$$\lambda_0 = 1, \quad \lambda_1 = e^{i\pi/2} = i, \quad \lambda_2 = e^{i\pi} = -1, \quad \text{and} \quad \lambda_3 = e^{i3\pi/2} = -i.$$

Notice that the eigenvalues are the fourth roots of unity—that is, solutions of the equation $\lambda^4 - 1 = 0$, where $\lambda^4 - 1$ is the characteristic polynomial of the matrix $\mathbf{C}$.

(a) (10 Points) Provide a well-labeled directed graph that corresponds to the connectivity matrix $\mathbf{C}$. The graph has four nodes. Each column of C represent a source node, and each row a sink node. Each node issues an edge to its next numbered node (with ν_4 issuing cyclically to ν_1). We have a cyclic graph



(b) (15 Points) *Without performing any matrix calculations or algebraic derivations,* determine the Google PageRank score vector for this network. Limit your explanation to preferably one, but at most two, sentences. No credit will be given for mathematical manipulations for this part.

By symmetry, no node should have a score different from the others, so $\underline{s}^* = \frac{1}{4}\mathbb{1} = \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$.

Note that $\underline{s}$ must be stochastic, hence the $\frac{1}{4}$ scaling.

FIRST Name: Larry LAST Name: PaigeRank SID (All Digits): 0123456789

E2.4 (Continued)

(c) (15 Points) Show that the eigenvector corresponding to the eigenvalue $\lambda_k = e^{ik\pi/2}$ can be expressed as

$$\boldsymbol{v}_k = \begin{bmatrix} 1 \\ \lambda_k^{-1} \\ \lambda_k^{-2} \\ \lambda_k^{-3} \end{bmatrix} = \begin{bmatrix} 1 \\ \lambda_k^{3} \\ \lambda_k^{2} \\ \lambda_k \end{bmatrix},$$

where we've scaled each eigenvector so that its top entry is 1. You need *not* show why that entry can never be zero.

Let $\underline{v}_k = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \Rightarrow C\underline{v}_k = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} d \\ a \\ b \\ c \end{bmatrix} \Rightarrow$

But $C\underline{v}_k = \lambda_k \underline{v}_k = \begin{bmatrix} \lambda_k a \\ \lambda_k b \\ \lambda_k c \\ \lambda_k d \end{bmatrix}$

$d = \lambda_k a$, $a = \lambda_k b$, $b = \lambda_k c$, and $c = \lambda_k d$.

The problem didn't ask you to show this—but rather merely assume it. None of the eigenvector entries can be zero, for otherwise it cascades into making all the other entries equal to zero — and this violates the definition of an eigenvector. So, assume $a = 1$. Then ...

$d = \lambda_k$; $1 = \lambda_k b \Rightarrow b = \lambda_k^{-1}$; $\lambda_k^{-1} = \lambda_k c \Rightarrow c = \lambda_k^{-2}$; and

$\lambda_k^{-2} = \lambda_k d \Rightarrow d = \lambda_k^{-3} = \lambda_k^{-3+4} = \lambda_k$ by the 4-periodicity of the complex exponentials $e^{ik\pi/2}$: $\lambda_k^{-3} = e^{-i3k\pi/2} = e^{-i3k\pi/2} e^{i4k\pi/2} = e^{ik\pi/2} = \lambda_k$

E2.4 (Continued)

(d) (25 Points) A population dynamics state-space model for this network is given by

$$\boldsymbol{s}[n+1] = \mathbf{C}\boldsymbol{s}[n], \qquad n = 0, 1, 2, \ldots$$

where $\boldsymbol{s}[n]$ denotes the state vector at sample n, which carries information about the population distribution across the nodes of the network. The initial state—that is, the initial distribution of the population across the nodes of the network—is $\boldsymbol{s}[0]$.

The state vector at sample n can be written as $\boldsymbol{s}[n] = \sum_{k=0}^{3} \alpha_k \lambda_k^n \boldsymbol{v}_k$.

(i) (10 Points) Show that the state vector is 4-periodic—that is $\boldsymbol{s}[n+4] = s[n]$ for all integers n.

$$\underline{s}[n] = \sum_{k=0}^{3} \alpha_k \underbrace{e^{ik\frac{\pi}{2}n}}_{\lambda_k^n} \underline{v}_k .$$

Note $\lambda_k^{n+4} = e^{ik\frac{\pi}{2}(n+4)} = e^{ik\frac{\pi}{2}n} e^{ik\frac{\pi}{2}4} = \lambda_k^n \Rightarrow \underline{s}[n+4] = \underline{s}[n] \ \forall n \in \mathbb{Z}$

$e^{ik2\pi} = 1$

(ii) (15 Points) Determine the average state over one period: $\boldsymbol{s}_{\text{AVG}} = \frac{1}{4}\sum_{n=0}^{3} \boldsymbol{s}[n]$.

$$\underline{s}_{AVG} = \frac{1}{4}\sum_{n=0}^{3} \underline{s}[n] = \frac{1}{4}\sum_{n=0}^{3}\sum_{k=0}^{3} \alpha_k \lambda_k^n \underline{v}_k = \sum_{k=0}^{3} \alpha_k \left(\frac{1}{4}\sum_{n=0}^{3} \lambda_k^n\right) \underline{v}_k$$

But $\sum_{n=0}^{3} \lambda_k^n = \sum_{n=0}^{3} e^{ik\frac{\pi}{2}n} = \begin{cases} 4 & \text{if } k=0 \\ 0 & \text{if } k \neq 0 \end{cases} \Rightarrow$

$$\sum_{n=0}^{3} \lambda_k^n = 4\delta[k] \Rightarrow \frac{1}{4}\sum_{n=0}^{3} \lambda_k^n = \delta[k] \Rightarrow \underline{s}_{AVG} = \sum_{k=0}^{3} \alpha_k \delta[k] \underline{v}_k$$

$\Rightarrow \underline{s}_{AVG} = \alpha_0 \underline{v}_0 = \alpha_0 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$. But $\mathbb{1}^T \underline{s}_{AVG} = \frac{1}{4}\sum_{n=0}^{3} \mathbb{1}^T \underline{s}[n] = \frac{1}{4} \cdot 4 = 1 \Rightarrow$

$\alpha_0 = 1/4 \Rightarrow \underline{s}_{AVG} = \frac{1}{4}\mathbb{1} = \underline{s}^*$ $\underline{s}_{AVG} = \underline{s}^*$ Google PageRank Score Vec.